

ALGORITHM

It is defined as a sequence of instructions that describe a method for solving a problem. In other words it is a step by step procedure for solving a problem.

Properties of Algorithms

1. Should be written in simple English
2. Each and every instruction should be precise and unambiguous.
3. Instructions in an algorithm should not be repeated infinitely.
4. Algorithm should conclude after a finite number of steps.
5. Should have an end point
6. Derived results should be obtained only after the algorithm terminates.

Qualities of a good algorithm

The following are the primary factors that are often used to judge the quality of the algorithms.

Time – To execute a program, the computer system takes some amount of time. The lesser is the time required, the better is the algorithm.

Memory – To execute a program, computer system takes some amount of memory space. The lesser is the memory required, the better is the algorithm.

Accuracy – Multiple algorithms may provide suitable or correct solutions to a given problem, some of these may provide more accurate results than others, and such algorithms may be suitable.

Example

Write an algorithm to print „Good Morning”

Step 1: Start

Step 2: Print “Good Morning”

Step 3: Stop

BUILDING BLOCKS OF ALGORITHMS (statements, state, control flow, functions)

Algorithms can be constructed from basic building blocks namely, sequence, selection and iteration.

Statements

Statement is a single action in a computer.

In a computer statements might include some of the following actions

1. input data-information given to the program
2. process data-perform operation on a given input
3. output data-processed result

State:

Transition from one process to another process under specified condition with in a time is called state.

Control flow:

The process of executing the individual statements in a given order is called control flow.

The control can be executed in three ways

1. Sequence
2. Selection
3. Iteration

Sequence:

All the instructions are executed one after another is called sequence execution.

Example:

Add two numbers:

Step 1: Start

Step 2: get a,b

Step 3: calculate $c=a+b$

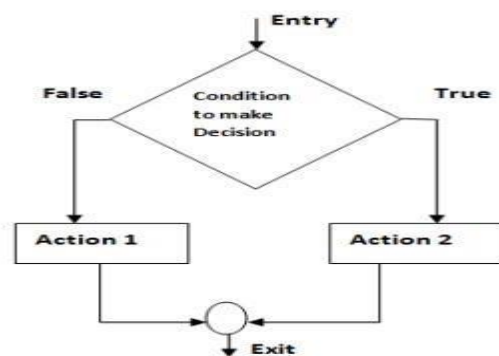
Step 4: Display c

Step 5: Stop

Selection:

A selection statement causes the program control to be transferred to a specific part of the program based upon the condition.

If the conditional test is true, one part of the program will be executed, otherwise it will execute the other part of the program.



Example

Write an algorithm to check whether he is eligible to vote?

Step 1: Start

Step 2: Get age

Step 3: if age ≥ 18 print "Eligible to vote"

Step 4: else print "Not eligible to vote"

Step 6: Stop

Iteration:

In some programs, certain set of statements are executed again and again based upon conditional test. i.e. executed more than one time. This type of execution is called looping or iteration.

Example

Write an algorithm to print all natural numbers up to n

Step 1: Start

Step 2: get n value.

Step 3: initialize $i=1$

Step 4: if $(i \leq n)$ go to step 5 else go to step 7

Step 5: Print i value and increment i value by 1

Step 6: go to step 4

Step 7: Stop

Functions:

1. Function is a sub program which consists of block of code (set of instructions) that performs a particular task.
2. For complex problems, the problem is been divided into smaller and simpler tasks during algorithm design.

Benefits of Using Functions

1. Reduction in line of code
2. code reuse
3. Better readability
4. Information hiding
5. Easy to debug and test
6. Improved maintainability

Example:

Algorithm for addition of two numbers using function

Main function()

Step 1: Start

Step 2: Call the function add()

Step 3: Stop

sub function add()

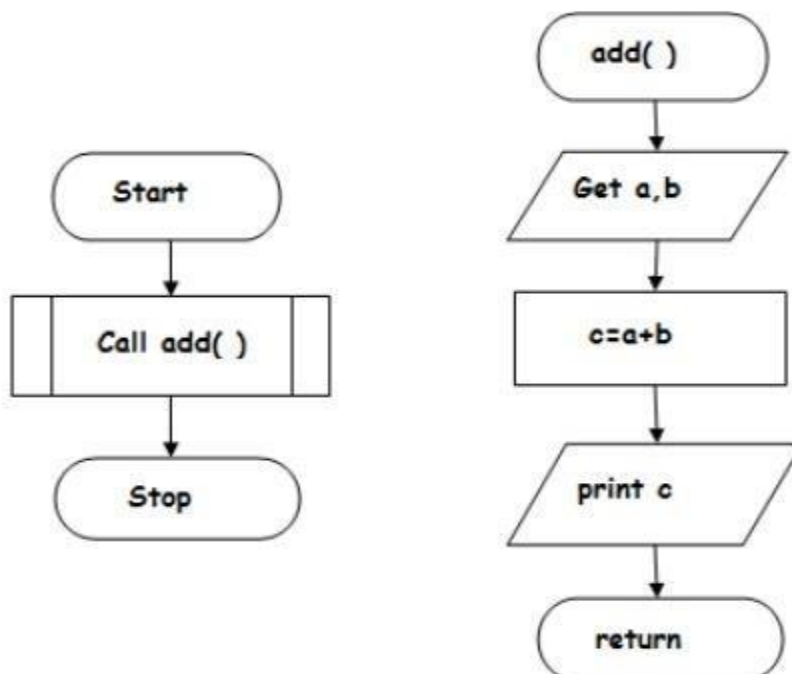
Step 1: Function start

Step 2: Get a, b Values

Step 3: add $c=a+b$

Step 4: Print c

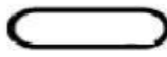
Step 5: Return



NOTATIONS

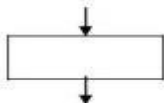
FLOW CHART

Flow chart is defined as graphical representation of the logic for problem solving. The purpose of flowchart is making the logic of the program clear in a visual representation.

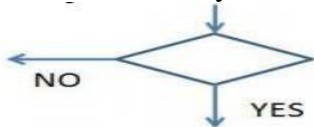
Symbol	Symbol Name	Description
	Flow Lines	Used to connect symbols
	Terminal	Used to start, pause or halt in the program logic
	Input/output	Represents the information entering or leaving the system
	Processing	Represents arithmetic and logical instructions
	Decision	Represents a decision to be made
	Connector	Used to Join different flow lines
	Sub function	used to call function

Rules for drawing a flowchart

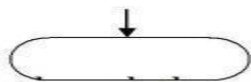
1. The flowchart should be clear, neat and easy to follow.
2. The flowchart must have a logical start and finish.
3. Only one flow line should come out from a process symbol.



4. Only one flow line should enter a decision symbol. However, two or three flow lines may leave the decision symbol.



5. Only one flow line is used with a terminal symbol.



6. Within standard symbols, write briefly and precisely.
7. Intersection of flow lines should be avoided.

Advantages of flowchart:

1. **Communication:** - Flowcharts are better way of communicating the logic of a system to all concerned.
2. **Effective analysis:** - With the help of flowchart, problem can be analyzed in more effective way.
3. **Proper documentation:** - Program flowcharts serve as a good program documentation, which is needed for various purposes.

4. **Efficient Coding:** - The flowcharts act as a guide or blueprint during the systems analysis and program development phase.
5. **Proper Debugging:** - The flowchart helps in debugging process.
6. **Efficient Program Maintenance:** - The maintenance of operating program becomes easy with the help of flowchart. It helps the programmer to put efforts more efficiently on that part.

Disadvantages of flow chart:

1. **Complex logic:** - Sometimes, the program logic is quite complicated. In that case, flowchart becomes complex and clumsy.
2. **Alterations and Modifications:** - If alterations are required the flowchart may require re-drawing completely.
3. **Reproduction:** - As the flowchart symbols cannot be typed, reproduction of flowchart becomes a problem.
4. **Cost:** For large application the time and cost of flowchart drawing becomes costly.

PSEUDO CODE:

1. Pseudo code consists of short, readable and formally styled English languages used for explain an algorithm.
2. It does not include details like variable declaration, subroutines.
3. It is easier to understand for the programmer or non programmer to understand the general working of the program, because it is not based on any programming language.
4. It gives us the sketch of the program before actual coding.
5. It is not a machine readable
6. Pseudo code can't be compiled and executed.
7. There is no standard syntax for pseudo code.

Guidelines for writing pseudo code:

1. Write one statement per line
2. Capitalize initial keyword
3. Indent to hierarchy
4. End multiline structure
5. Keep statements language independent

Common keywords used in pseudo code

The following gives common keywords used in pseudo codes.

1. **//:** This keyword used to represent a comment.
2. **BEGIN,END:** Begin is the first statement and end is the last statement.
3. **INPUT, GET, READ:** The keyword is used to inputting data.
4. **COMPUTE, CALCULATE:** used for calculation of the result of the given expression.
5. **ADD, SUBTRACT, INITIALIZE** used for addition, subtraction and initialization.
6. **OUTPUT, PRINT, DISPLAY:** It is used to display the output of the program.
7. **IF, ELSE, ENDIF:** used to make decision.
8. **WHILE, ENDWHILE:** used for iterative statements.
9. **FOR, ENDFOR:** Another iterative incremented/decremented tested automatically.

Syntax for if else:

```
IF (condition) THEN
statement
...
ELSE
statement
```

```
...
ENDIF
```

Example: Greates of two numbers

```
BEGIN
READ a,b
IF (a>b) THEN
DISPLAY a is greater
ELSE
DISPLAY b is greater
END IF
END
```

Syntax for For:

```
FOR( start-value to end-value) DO
statement
...
ENDFOR
```

Example: Print n natural numbers

```
BEGIN
GET n
INITIALIZE i=1
FOR (i<=n) DO
PRINT i
i=i+1
ENDFOR
END
```

Syntax for While:

```
WHILE (condition) DO
statement
...
ENDWHILE
```

Example: Print n natural numbers

```
BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
PRINT i
i=i+1
ENDWHILE
END
```

Advantages:

1. Pseudo is independent of any language; it can be used by most programmers.
2. It is easy to translate pseudo code into a programming language.
3. It can be easily modified as compared to flowchart.
4. Converting a pseudo code to programming language is very easy as compared with converting a flowchart to programming language.

Disadvantages:

1. It does not provide visual representation of the program's logic.
2. There are no accepted standards for writing pseudo codes.
3. It cannot be compiled nor executed.
4. For a beginner, It is more difficult to follow the logic or write pseudo code as compared to flowchart.

Example:**Addition of two numbers:**

```

BEGIN
GET a,b
ADD c=a+b
PRINT c
END

```

Algorithm	Flowchart	Pseudo code
An algorithm is a sequence of instructions used to solve a problem	It is a graphical representation of algorithm	It is a language representation of algorithm.
User needs knowledge to write algorithm.	not need knowledge of program to draw or understand flowchart	Not need knowledge of program language to understand or write a pseudo code.

PROGRAMMING LANGUAGE

A programming language is a set of symbols and rules for instructing a computer to perform specific tasks. The programmers have to follow all the specified rules before writing program using programming language. The user has to communicate with the computer using language which it can understand.

Types of programming language

1. Machine language
2. Assembly language
3. High level language

Machine language:

The computer can understand only machine language which uses 0's and 1's. In machine language the different instructions are formed by taking different combinations of 0's and 1's.

Advantages:**Translation free:**

Machine language is the only language which the computer understands. For executing any program written in any programming language, the conversion to machine language is necessary. The program written in machine language can be executed directly on computer. In this case any conversion process is not required.

High speed

The machine language program is translation free. Since the conversion time is saved, the execution of machine language program is extremely fast.

Disadvantage:

1. It is hard to find errors in a program written in the machine language.
2. Writing program in machine language is a time consuming process.

Machine dependent: According to architecture used, the computer differs from each other. So machine language differs from computer to computer. So a program developed for a particular type of computer may not run on other type of computer.

Assembly language:

To overcome the issues in programming language and make the programming process easier, an assembly language is developed which is logically equivalent to machine language but it is easier for people to read, write and understand.

1. Assembly language is symbolic representation of machine language. Assembly languages are symbolic programming language that uses symbolic notation to represent machine language instructions. They are called low level language because they are so closely related to the machines.

Assembler

Assembler is the program which translates assembly language instruction in to a machine language.

1. Easy to understand and use.
2. It is easy to locate and correct errors.

Disadvantage

Machine dependent

The assembly language program which can be executed on the machine depends on the architecture of that computer.

Hard to learn

It is machine dependent, so the programmer should have the hardware knowledge to create applications using assembly language.

Less efficient

1. Execution time of assembly language program is more than machine language program.
2. Because assembler is needed to convert from assembly language to machine language.

High level language

High level language contains English words and symbols. The specified rules are to be followed while writing program in high level language. The interpreter or compilers are used for converting these programs in to machine readable form.

Translating high level language to machine language

The programs that translate high level language in to machine language are called interpreter or compiler.

Compiler:

A compiler is a program which translates the source code written in a high level language in to object code which is in machine language program. Compiler reads the whole program written in high level language and translates it to machine language. If any error is found it display error message on the screen.

Interpreter

Interpreter translates the high level language program in line by line manner. The interpreter translates a high level language statement in a source program to a machine code and executes it immediately before translating the next statement. When an error is found the execution of the program is halted and error message is displayed on the screen.

Advantages**Readability**

High level language is closer to natural language so they are easier to learn and understand

Machine independent

High level language program have the advantage of being portable between machines.

Easy debugging

Easy to find and correct error in high level language

Disadvantages**Less efficient**

The translation process increases the execution time of the program. Programs in high level language require more memory and take more execution time to execute.

They are divided into following categories:

- | | |
|--------------------------------------|---|
| 1. Interpreted programming languages | 2. Functional programming languages |
| 3. Compiled programming languages | 4. Procedural programming languages |
| 5. Scripting programming language | 6. Markup programming language |
| 7. Concurrent programming language | 8. Object oriented programming language |

Interpreted programming languages:

An interpreted language is a programming language for which most of its implementation executes instructions directly, without previously compiling a program into machine language instructions. The interpreter executes the program directly translating each statement into a sequence of one or more subroutines already compiled into machine code.

Examples:

Pascal, Python

Functional programming language:

Functional programming language defines every computation as a mathematical evaluation. They focus on the programming languages are bound to mathematical calculations

Examples:

Clean, Haskell

Compiled Programming language:

A compiled programming is a programming language whose implementation are typically compilers and not interpreters.

It will produce a machine code from source code.

Examples:

C, C++, C#, JAVA

Procedural programming language:

Procedural (imperative) programming implies specifying the steps that the programs should take to reach to an intended state.

A procedure is a group of statements that can be referred through a procedure call. Procedures help in the reuse of code. Procedural programming makes the programs structured and easily traceable for program flow.

Examples:

Hyper talk, MATLAB

Scripting language:

Scripting language are programming languages that control an application. Scripts can execute independent of any other application. They are mostly embedded in the application that they control and are used to automate frequently executed tasks like communicating with external program.

Examples:

Apple script, VB script

Markup languages:

A markup language is an artificial language that uses annotations to text that define hoe the text is to be displayed.

Examples:

HTML, XML

Concurrent programming language:

Concurrent programming is a computer programming technique that provides for the execution of operation concurrently, either with in a single computer or across a number of systems.

Examples:

Joule, Limbo

Object oriented programming language:

Object oriented programming is a programming paradigm based on the concept of objects which may contain data in the form of procedures often known as methods.

Examples:

Lava, Moto

ALGORITHMIC PROBLEM SOLVING:

Algorithmic problem solving is solving problem that require the formulation of an algorithm for the solution.

Understanding the Problem

1. It is the process of finding the input of the problem that the algorithm solves.
2. It is very important to specify exactly the set of inputs the algorithm needs to handle.
3. A correct algorithm is not one that works most of the time, but one that works correctly for *all* legitimate inputs.

Ascertaining the Capabilities of the Computational Device

1. If the instructions are executed one after another, it is called sequential algorithm.
2. If the instructions are executed concurrently, it is called parallel algorithm.

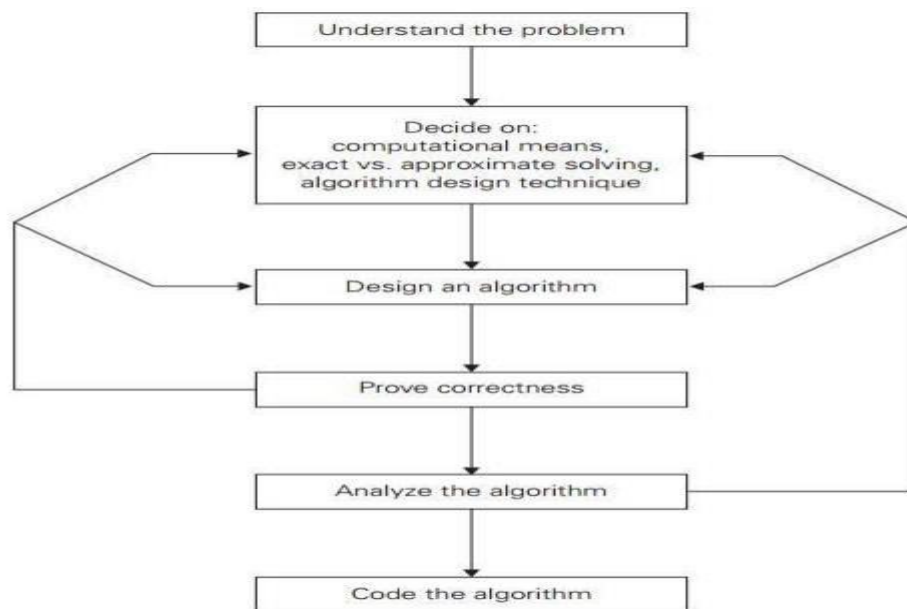


FIGURE 1.2 Algorithm design and analysis process.

Choosing between Exact and Approximate Problem Solving

1. The next principal decision is to choose between solving the problem exactly or solving it approximately.
2. Based on this, the algorithms are classified as *exact algorithm* and *approximation algorithm*.

Deciding a data structure:

1. Data structure plays a vital role in designing and analysis the algorithms.
2. Some of the algorithm design techniques also depend on the structuring data specifying a problem's instance
3. Algorithm+ Data structure=programs.

Algorithm Design Techniques

1. An **algorithm design technique** (or “strategy” or “paradigm”) is a general approach to solving problems algorithmically that is applicable to a variety of problems from different areas of computing.
2. Learning these techniques is of utmost importance for the following reasons.
3. First, they provide guidance for designing algorithms for new problems,
4. Second, algorithms are the cornerstone of computer science

Methods of Specifying an Algorithm

1. **Pseudocode** is a mixture of a natural language and programming language-like constructs. Pseudocode is usually more precise than natural language, and its usage often yields more succinct algorithm descriptions.
2. In the earlier days of computing, the dominant vehicle for specifying algorithms was a **flowchart**, a method of expressing an algorithm by a collection of connected geometric shapes containing descriptions of the algorithm's steps.
3. **Programming language** can be fed into an electronic computer directly. Instead, it needs to be converted into a computer program written in a particular computer language. We can look at such a program as yet another way of
4. specifying the algorithm, although it is preferable to consider it as the algorithm's implementation.

Proving an Algorithm's Correctness

1. Once an algorithm has been specified, you have to prove its **correctness**. That is, you have to prove that the algorithm yields a required result for every legitimate input in a finite amount of time.

2. A common technique for proving correctness is to use mathematical induction because an algorithm's iterations provide a natural sequence of steps needed for such proofs.
3. It might be worth mentioning that although tracing the algorithm's performance for a few specific inputs can be a very worthwhile activity, it cannot prove the algorithm's correctness conclusively. But in order to show that an algorithm is incorrect, you need just one instance of its input for which the algorithm fails.

Analysing an Algorithm

1. Efficiency.

Time efficiency, indicating how fast the algorithm runs,

Space efficiency, indicating how much extra memory it uses.

2. simplicity.

1. An algorithm should be precisely defined and investigated with mathematical expressions.
2. Simpler algorithms are easier to understand and easier to program.
3. Simple algorithms usually contain fewer bugs.

Coding an Algorithm

1. Most algorithms are destined to be ultimately implemented as computer programs. Programming an algorithm presents both a peril and an opportunity.
2. A working program provides an additional opportunity in allowing an empirical analysis of the underlying algorithm. Such an analysis is based on timing the program on several inputs and then analysing the results obtained.

SIMPLE STRATEGIES FOR DEVELOPING ALGORITHMS:

1. Iterations
2. Recursions

1. Iterations:

A sequence of statements is executed until a specified condition is true is called iterations.

1. for loop
2. While loop

Syntax for For:

FOR(*start-value to end-value*) DO

Statement

...

ENDFOR

Example: Print n natural numbers

BEGIN

GET n

INITIALIZE i=1

FOR (i<=n) DO

PRINT i

i=i+1

ENDFOR

END

Syntax for While:

WHILE (condition) DO

Statement

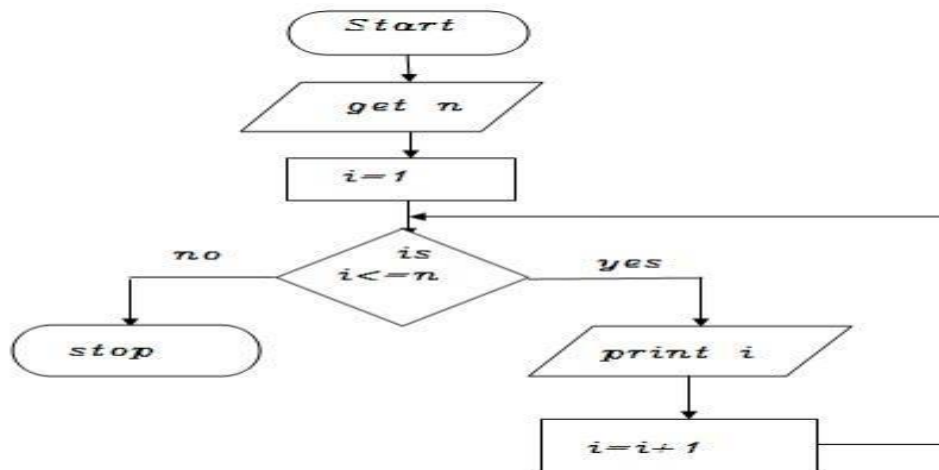
...

ENDWHILE

Example: Print n natural numbers

```

BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
PRINT i
i=i+1
ENDWHILE
END
    
```



Recursions:

1. A function that calls itself is known as recursion.
2. Recursion is a process by which a function calls itself repeatedly until some specified condition has been satisfied.

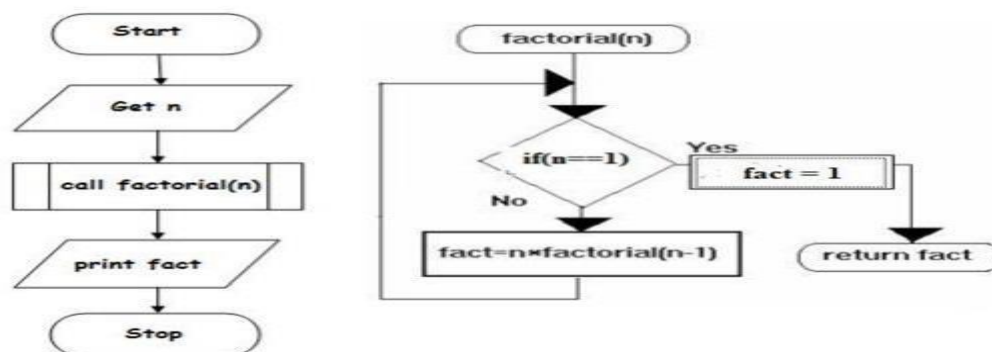
Algorithm for factorial of n numbers using recursion:

Main function:

- Step1: Start
- Step2: Get n
- Step3: call factorial(n)
- Step4: print fact
- Step5: Stop

Sub function factorial(n):

- Step1: if(n==1) then fact=1 return fact
- Step2: else fact=n*factorial(n-1) and return fact



Pseudo code for factorial using recursion:

Main function:

```
BEGIN
GET n
CALL factorial(n)
PRINT fact
BIN
```

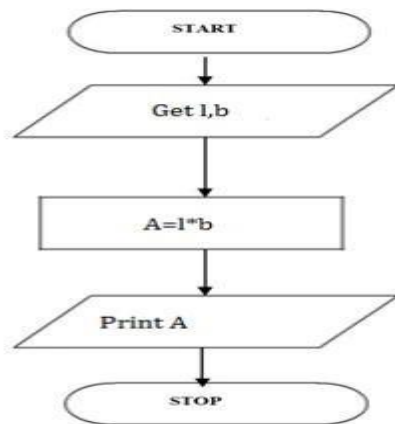
Sub function factorial(n):

```
IF(n==1) THEN
    fact=1
    RETURN fact
ELSE
    RETURN fact=n*factorial(n-1)
```

More examples:

Write an algorithm to find area of a rectangle

Step 1: Start
 Step 2: get l,b values
 Step 3: Calculate $A=l*b$
 Step 4: Display A
 Step 5: Stop



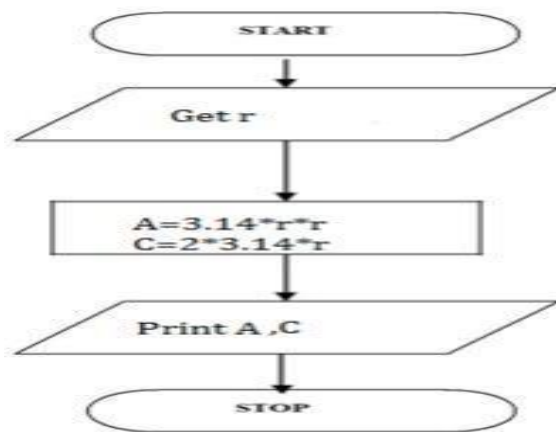
```
BEGIN
READ l,b
CALCULATE A=l*b
DISPLAY A
END
```

Write an algorithm for Calculating area and circumference of circle

Step 1: Start
 Step 2: get r value
 Step 3: Calculate $A=3.14*r*r$
 Step 4: Calculate $C=2*3.14*r$
 Step 5: Display A,C
 Step 6: Stop

```
BEGIN
READ r
CALCULATE A and C
A=3.14*r*r
C=2*3.14*r
DISPLAY A
```

END



Write an algorithm for Calculating simple interest

Step 1: Start

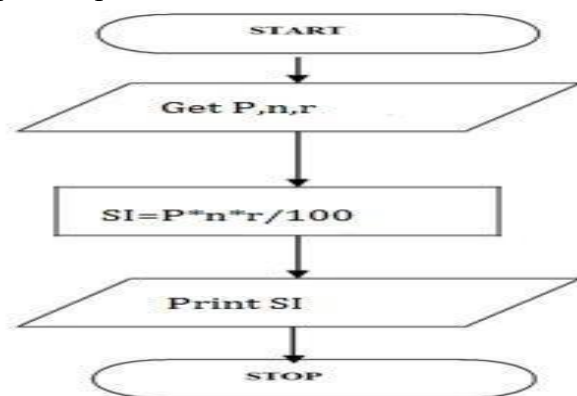
Step 2: get P, n, r value

Step3: Calculate

$$SI = (p * n * r) / 100$$

Step 4: Display S

Step 5: Stop



BEGIN

READ P, n, r

CALCULATE S

$$SI = (p * n * r) / 100$$

DISPLAY SI

END

Write an algorithm for Calculating engineering cutoff

Step 1: Start

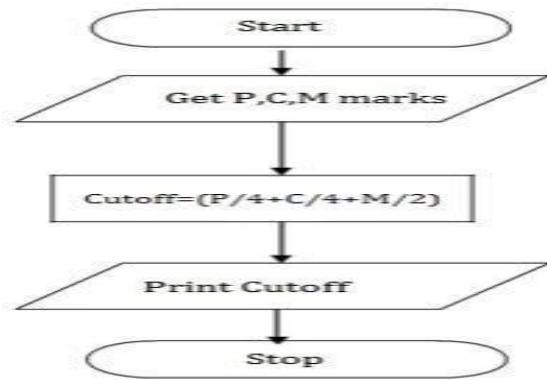
Step2: get P,C,M value

Step3: calculate

$$\text{Cutoff} = (P/4 + C/4 + M/2)$$

Step 4: Display Cutoff

Step 5: Stop

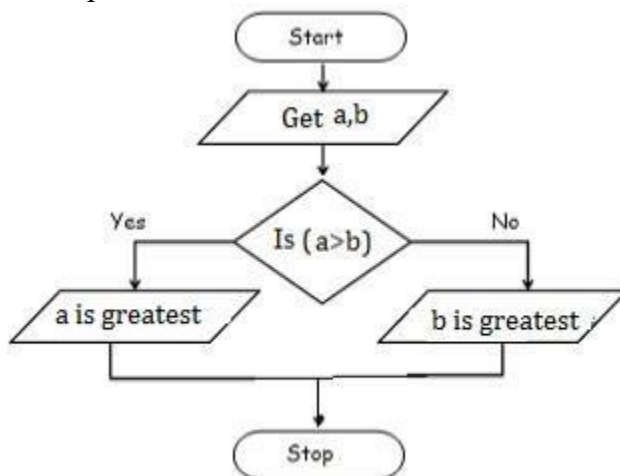


```

BEGIN
READ P,C,M
CALCULATE
Cutoff= (P/4+C/4+M/2)
DISPLAY Cutoff
END
    
```

To check greatest of two numbers

Step 1: Start
 Step 2: get a,b value
 Step 3: check if(a>b) print a is greater
 Step 4: else b is greater
 Step 5: Stop

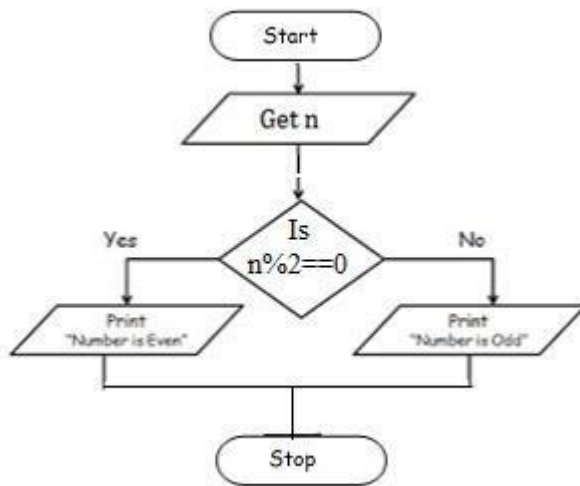


```

BEGIN
READ a,b
IF (a>b) THEN
DISPLAY a is greater
ELSE
DISPLAY b is greater
END IF
END
    
```

To check leap year or not

Step 1: Start
 Step 2: get y
 Step 3: if(y%4==0) print leap year
 Step 4: else print not leap year
 Step 5: Stop

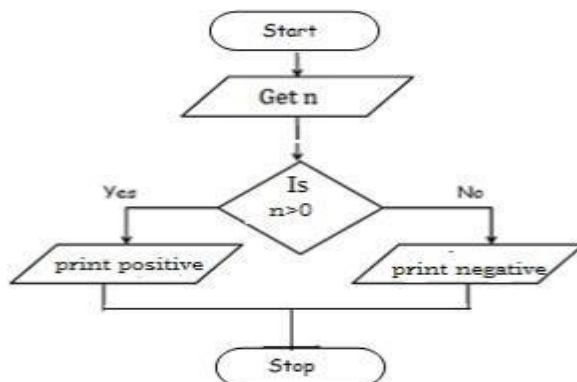


```

BEGIN
READ y
IF (y%4==0) THEN
DISPLAY leap year
ELSE
DISPLAY not leap year
END IF
END
  
```

To check positive or negative number

Step 1: Start
Step 2: get num
Step 3: check if(num>0) print a is positive
Step 4: else num is negative
Step 5: Stop



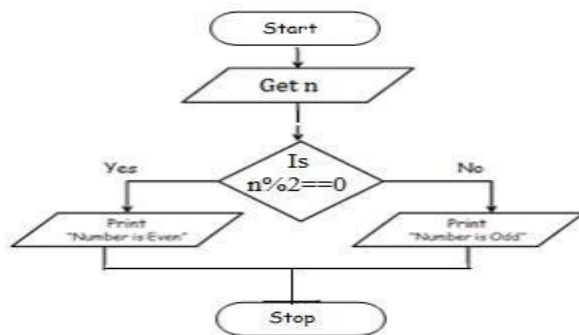
```

BEGIN
READ num
IF (num>0) THEN
DISPLAY num is positive
ELSE
DISPLAY num is negative
END IF
END
  
```

To check odd or even number

Step 1: Start

Step 2: get num
 Step 3: check if($\text{num} \% 2 == 0$) print num is even
 Step 4: else num is odd
 Step 5: Stop

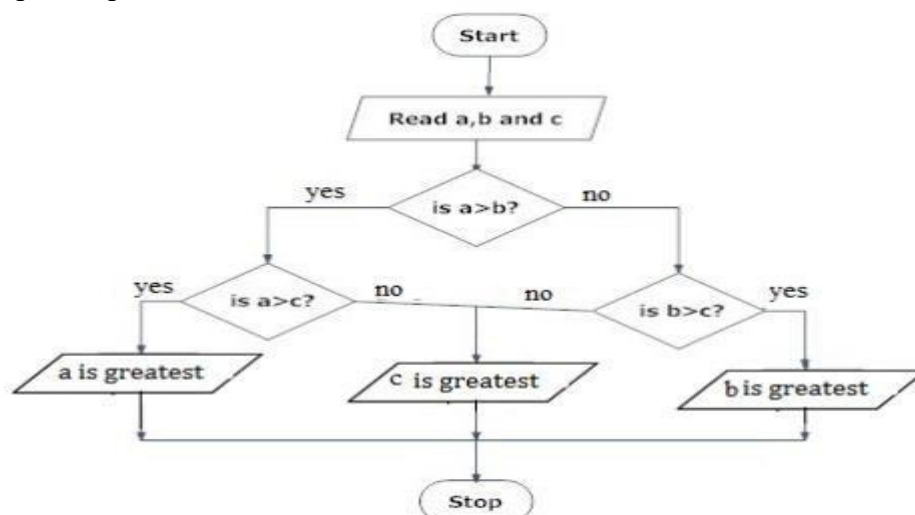


```

BEGIN
READ num
IF (num%2==0) THEN
DISPLAY num is even
ELSE
DISPLAY num is odd
END IF
END
    
```

To check greatest of three numbers

Step1: Start
 Step2: Get A, B, C
 Step3: if($A > B$) goto Step4 else goto step5
 Step4: If($A > C$) print A else print C
 Step5: If($B > C$) print B else print C
 Step6: Stop



```

BEGIN
READ a, b, c
IF (a > b) THEN
IF(a > c) THEN
DISPLAY a is greater
ELSE
DISPLAY c is greater
    
```

```

END IF
ELSE
IF(b>c) THEN
DISPLAY b is greater
ELSE
DISPLAY c is greater
END IF
END IF
END

```

Write an algorithm to check whether given number is +ve, -ve or zero.

Step 1: Start

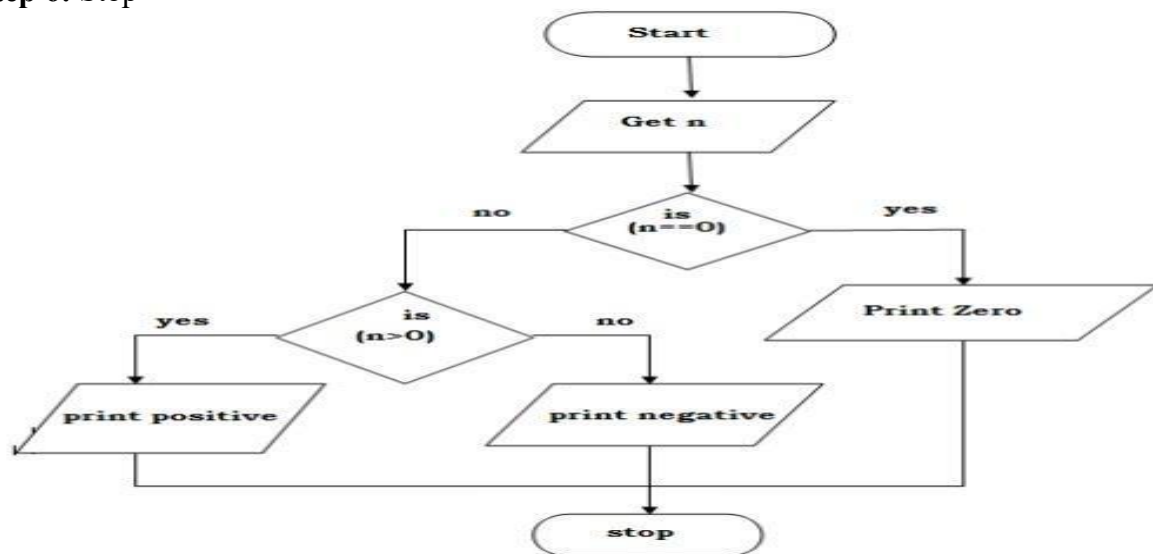
Step 2: Get n value.

Step 3: if (n ==0) print “Given number is Zero” Else goto step4

Step 4: if (n > 0) then Print “Given number is +ve”

Step 5: else Print “Given number is -ve”

Step 6: Stop



```

BEGIN
GET n
IF(n==0) THEN
    DISPLAY “ n is zero”
ELSE
    IF(n>0) THEN
        DISPLAY “n is positive”
    ELSE
        DISPLAY “n is positive”
    END IF
END IF
END IF
END

```

Write an algorithm to print all natural numbers up to n

Step 1: Start

Step 2: get n value.

Step 3: initialize i=1

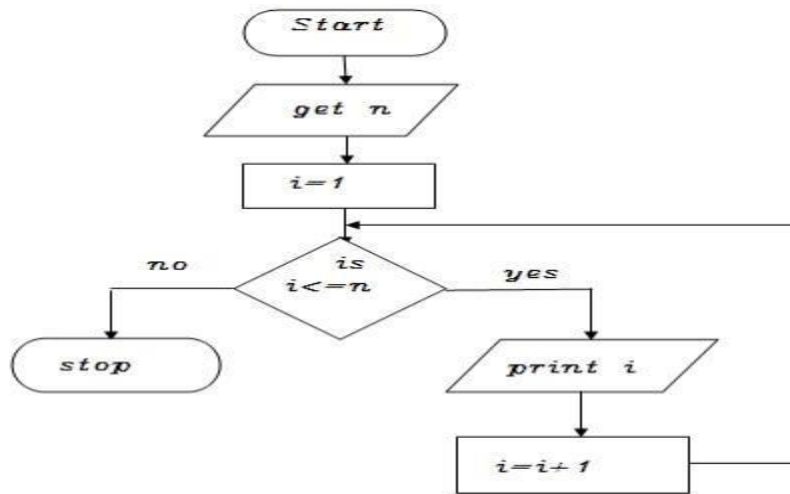
Step 4: if (i<=n) go to step 5 else go to step 8

Step 5: Print i value

step 6 : increment i value by 1

Step 7: go to step 4

Step 8: Stop

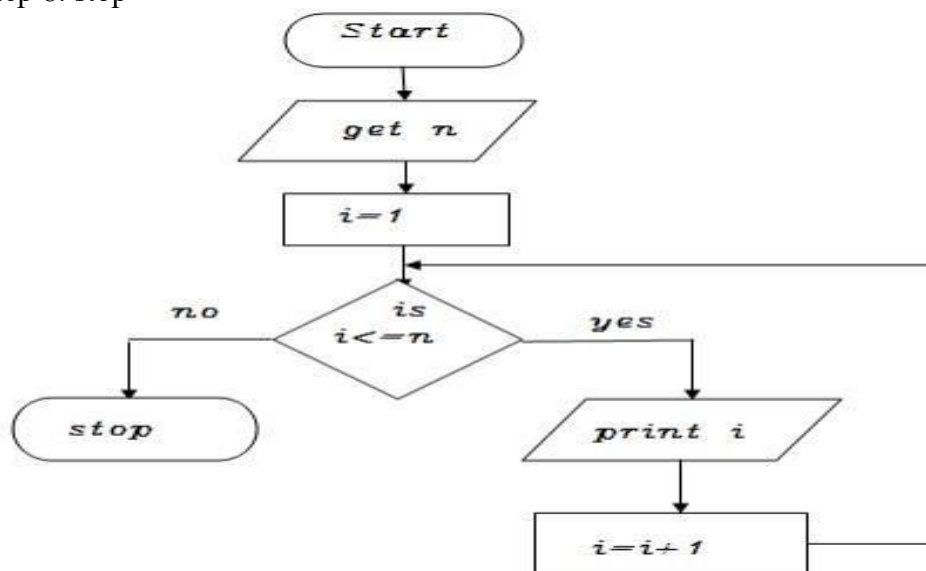


```

BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
PRINT i
i=i+1
ENDWHILE
END
    
```

Write an algorithm to print n odd numbers

Step 1: start
 step 2: get n value
 step 3: set initial value i=1
 step 4: check if(i<=n) goto step 5 else goto step 8
 step 5: print i value
 step 6: increment i value by 2
 step 7: goto step 4
 step 8: stop



```

BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
    PRINT i
ENDWHILE
END
    
```

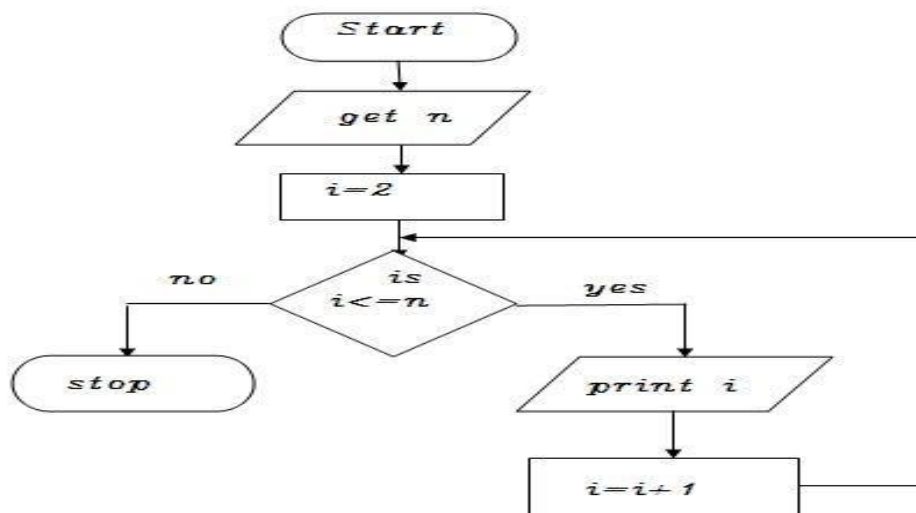
```

    i=i+2
ENDWHILE
END

```

Write an algorithm to print n even numbers

Step 1: start
 step 2: get n value
 step 3: set initial value $i=2$
 step 4: check if($i \leq n$) goto step 5 else goto step8
 step 5: print i value
 step 6: increment i value by 2
 step 7: goto step 4
 step 8: stop



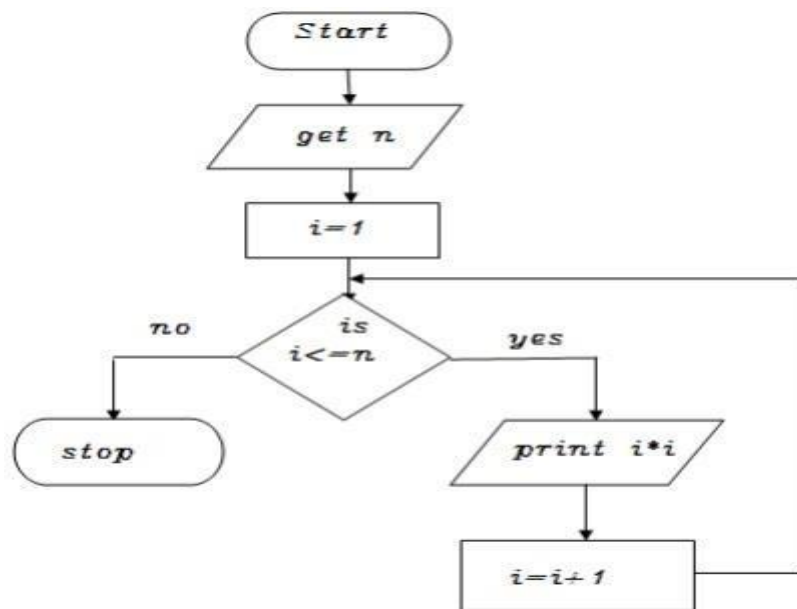
```

BEGIN
GET n
INITIALIZE i=2
WHILE(i<=n) DO
    PRINT i
    i=i+2
ENDWHILE
END

```

Write an algorithm to print squares of a number

Step 1: start
 step 2: get n value
 step 3: set initial value $i=1$
 step 4: check i value if($i \leq n$) goto step 5 else goto step8
 step 5: print $i*i$ value
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: stop

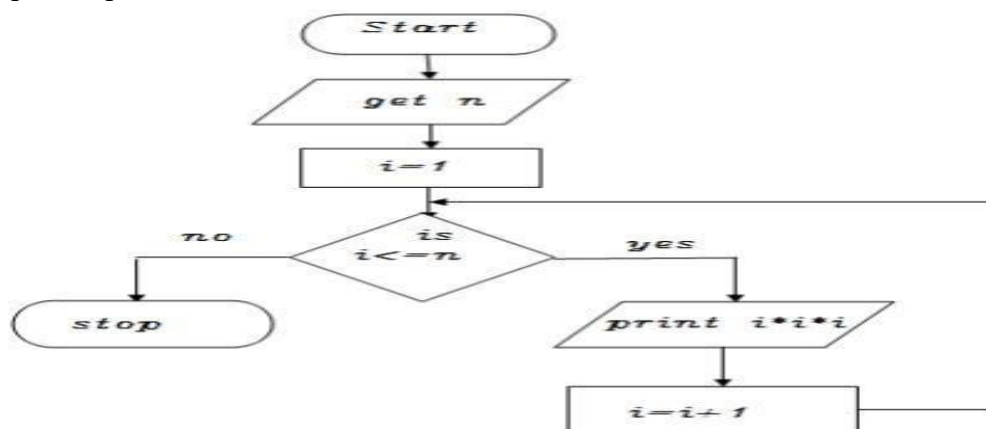


```

BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
    PRINT i*i
    i=i+1
ENDWHILE
END
  
```

Write an algorithm to print to print cubes of a number

Step 1: start
 step 2: get n value
 step 3: set initial value i=1
 step 4: check i value if(i<=n) goto step 5 else goto step8
 step 5: print i*i *i value
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: stop



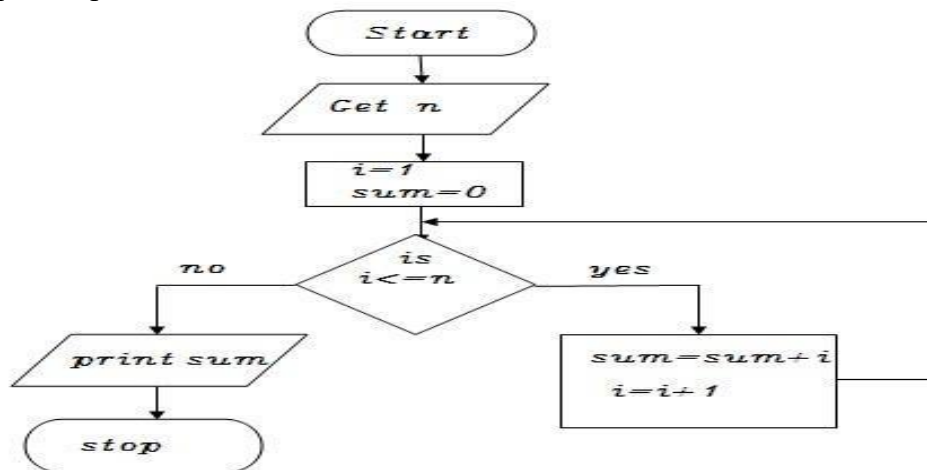
```

BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
    PRINT i*i*i
    i=i+1
ENDWHILE
  
```

END

Write an algorithm to find sum of a given number

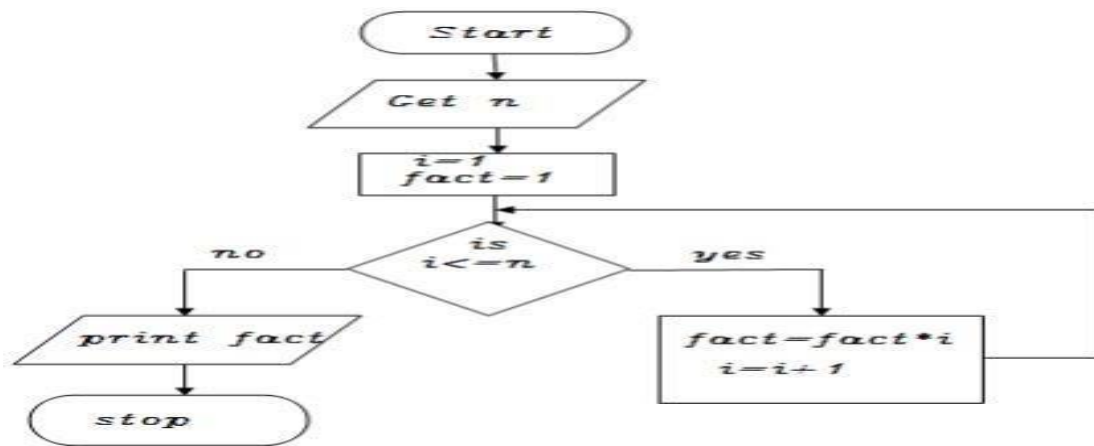
Step 1: start
 step 2: get n value
 step 3: set initial value $i=1$, $sum=0$
 Step 4: check i value if($i \leq n$) goto step 5 else goto step8
 step 5: calculate $sum=sum+i$
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: print sum value
 step 9: stop



BEGIN
 GET n
 INITIALIZE $i=1, sum=0$
 WHILE($i \leq n$) DO
 $sum=sum+i$
 $i=i+1$
 ENDWHILE
 PRINT sum
 END

Write an algorithm to find factorial of a given number

Step 1: start
 step 2: get n value
 step 3: set initial value $i=1$, $fact=1$
 Step 4: check i value if($i \leq n$) goto step 5 else goto step8
 step 5: calculate $fact=fact*i$
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: print fact value
 step 9: stop

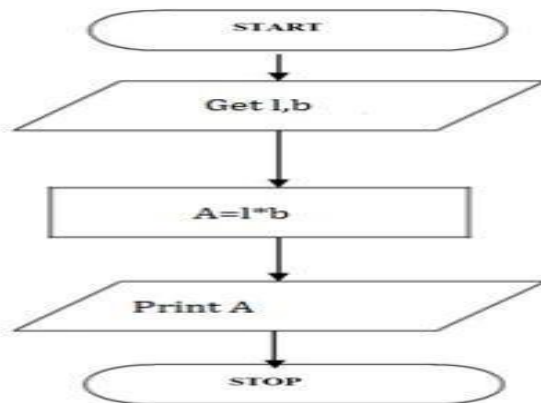


```

BEGIN
GET n
INITIALIZE i=1,fact=1
WHILE(i<=n) DO
    fact=fact*i
    i=i+1
ENDWHILE
PRINT fact
END
    
```

Write an algorithm to find area of a rectangle

Step 1: Start
 Step 2: get l,b values
 Step 3: Calculate $A=l*b$
 Step 4: Display A
 Step 5: Stop

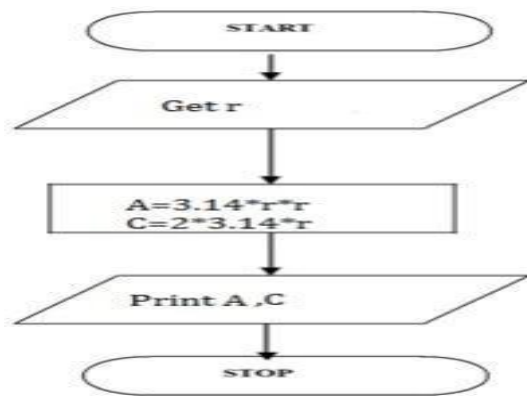


```

BEGIN
READ l,b
CALCULATE A=l*b
DISPLAY A
END
    
```

Write an algorithm for Calculating area and circumference of circle

Step 1: Start
 Step 2: get r value
 Step 3: Calculate $A=3.14*r*r$
 Step 4: Calculate $C=2.3.14*r$
 Step 5: Display A,C
 Step 6: Stop

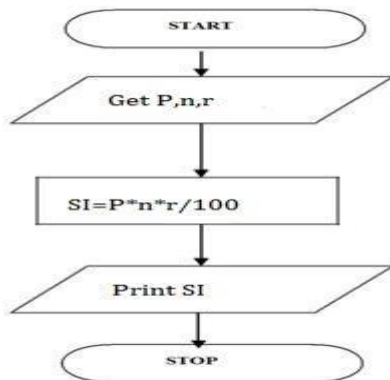


```

BEGIN
READ r
CALCULATE A and C
A=3.14*r*r
C=2*3.14*r
DISPLAY A
END
    
```

Write an algorithm for Calculating simple interest

Step 1: Start
 Step 2: get P, n, r value
 Step3: Calculate
 $SI = (p*n*r)/100$
 Step 4: Display S
 Step 5: Stop

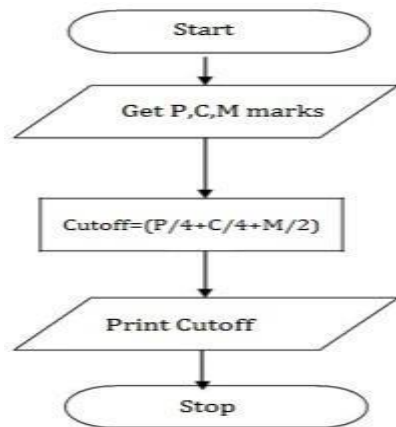


```

BEGIN
READ P, n, r
CALCULATE S
SI=(p*n*r)/100
DISPLAY SI
END
    
```

Write an algorithm for Calculating engineering cutoff

Step 1: Start
 Step2: get P,C,M value
 Step3: calculate
 $Cutoff = (P/4 + C/4 + M/2)$
 Step 4: Display Cutoff
 Step 5: Stop

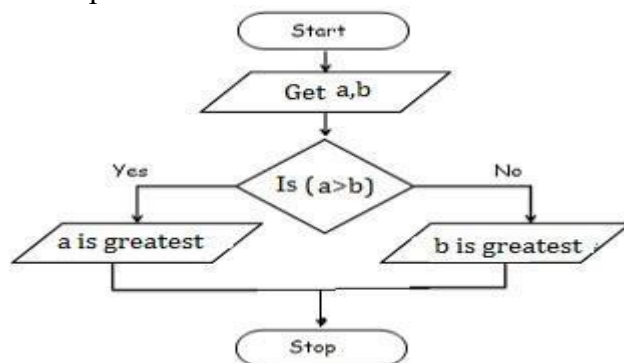


```

BEGIN
READ P,C,M
CALCULATE
Cutoff= (P/4+C/4+M/2)
DISPLAY Cutoff
END
  
```

To check greatest of two numbers

Step 1: Start
 Step 2: get a,b value
 Step 3: check if(a>b) print a is greater
 Step 4: else b is greater
 Step 5: Stop

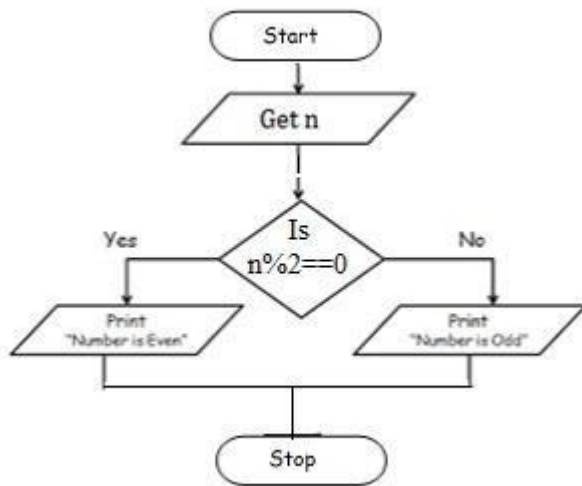


```

BEGIN
READ a,b
IF (a>b) THEN
DISPLAY a is greater
ELSE
DISPLAY b is greater
END IF
END
  
```

To check leap year or not

Step 1: Start
 Step 2: get y
 Step 3: if(y%4==0) print leap year
 Step 4: else print not leap year
 Step 5: Stop

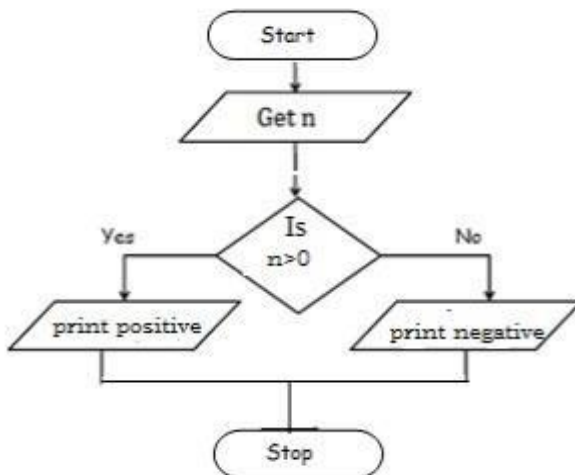


```

BEGIN
READ y
IF (y%4==0) THEN
DISPLAY leap year
ELSE
DISPLAY not leap year
END IF
END
  
```

To check positive or negative number

Step 1: Start
Step 2: get num
Step 3: check if(num>0) print a is positive
Step 4: else num is negative
Step 5: Stop

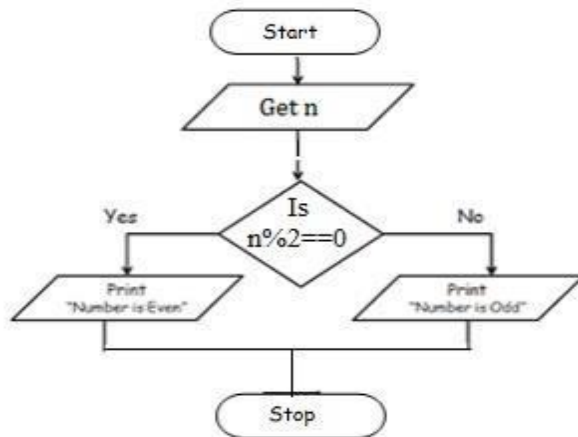


```

BEGIN
READ num
IF (num>0) THEN
DISPLAY num is positive
ELSE
DISPLAY num is negative
END IF
END
  
```

To check odd or even number

Step 1: Start
 Step 2: get num
 Step 3: check if($\text{num} \% 2 == 0$) print num is even
 Step 4: else num is odd
 Step 5: Stop

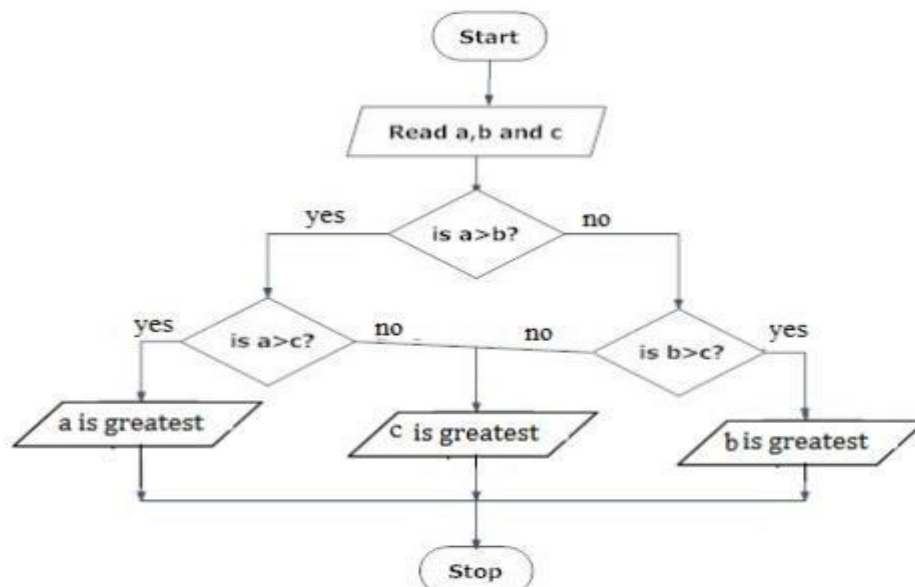


```

BEGIN
READ num
IF (num%2==0) THEN
DISPLAY num is even
ELSE
DISPLAY num is odd
END IF
END
    
```

To check greatest of three numbers

Step1: Start
 Step2: Get A, B, C
 Step3: if($A > B$) goto Step4 else goto step5
 Step4: If($A > C$) print A else print C
 Step5: If($B > C$) print B else print C
 Step6: Stop



```

BEGIN
READ a, b, c
IF (a > b) THEN
IF(a > c) THEN
    
```

```

DISPLAY a is greater
ELSE
DISPLAY c is greater
END IF
ELSE
IF(b>c) THEN
DISPLAY b is greater
ELSE
DISPLAY c is greater
END IF
END IF
END
    
```

Write an algorithm to check whether given number is +ve, -ve or zero.

Step 1: Start

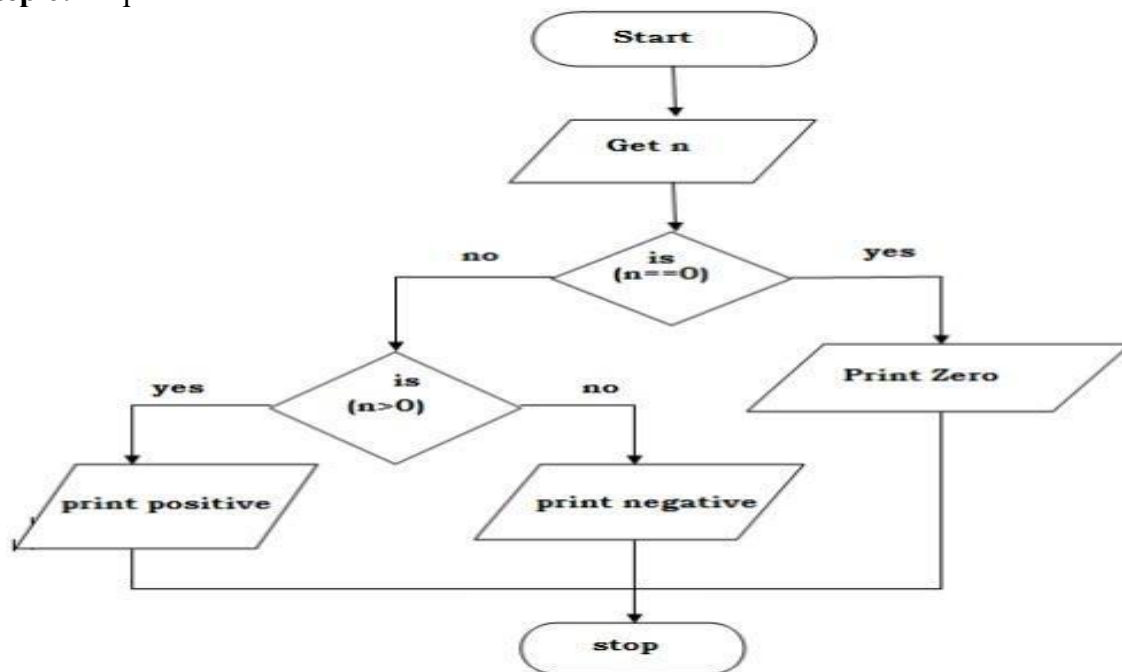
Step 2: Get n value.

Step 3: if (n==0) print “Given number is Zero” Else goto step4

Step 4: if (n > 0) then Print “Given number is +ve”

Step 5: else Print “Given number is -ve”

Step 6: Stop



```

BEGIN
GET n
IF(n==0) THEN
    DISPLAY “ n is zero”
ELSE
    IF(n>0) THEN
        DISPLAY “n is positive”
    ELSE
        DISPLAY “n is positive”
    END IF
END IF
END IF
END
    
```

Write an algorithm to print all natural numbers up to n

Step 1: Start

Step 2: get n value.

Step 3: initialize $i=1$

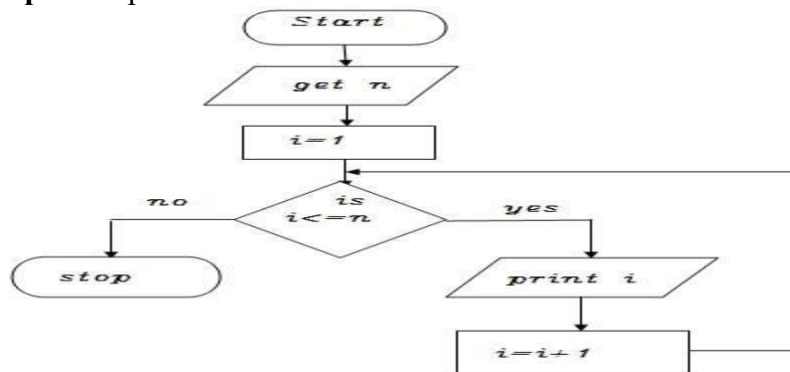
Step 4: if $(i \leq n)$ go to step 5 else go to step 8

Step 5: Print i value

step 6 : increment i value by 1

Step 7: go to step 4

Step 8: Stop



BEGIN

GET n

INITIALIZE $i=1$

WHILE($i \leq n$) DO

PRINT i

$i=i+1$

ENDWHILE

END

Write an algorithm to print n odd numbers

Step 1: start

step 2: get n value

step 3: set initial value $i=1$

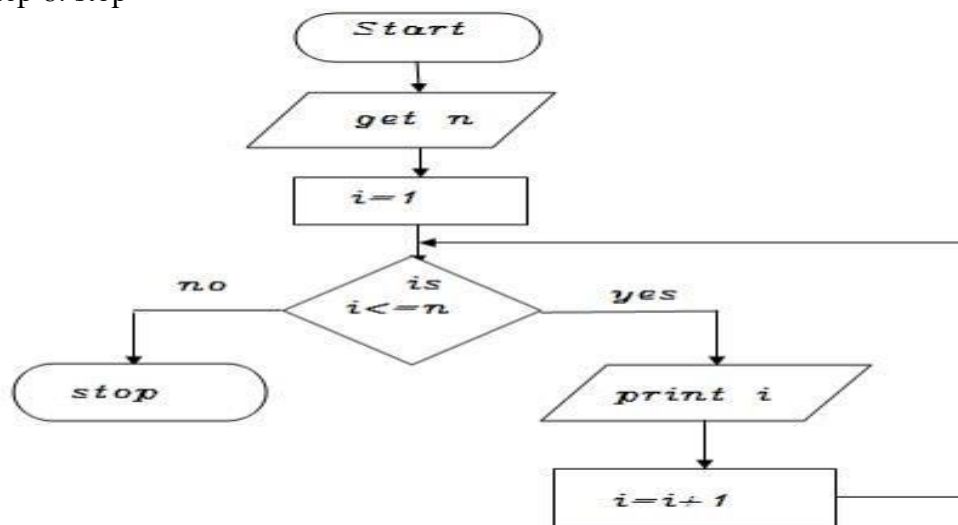
step 4: check if $(i \leq n)$ goto step 5 else goto step 8

step 5: print i value

step 6: increment i value by 2

step 7: goto step 4

step 8: stop



BEGIN

GET n

INITIALIZE $i=1$

WHILE($i \leq n$) DO

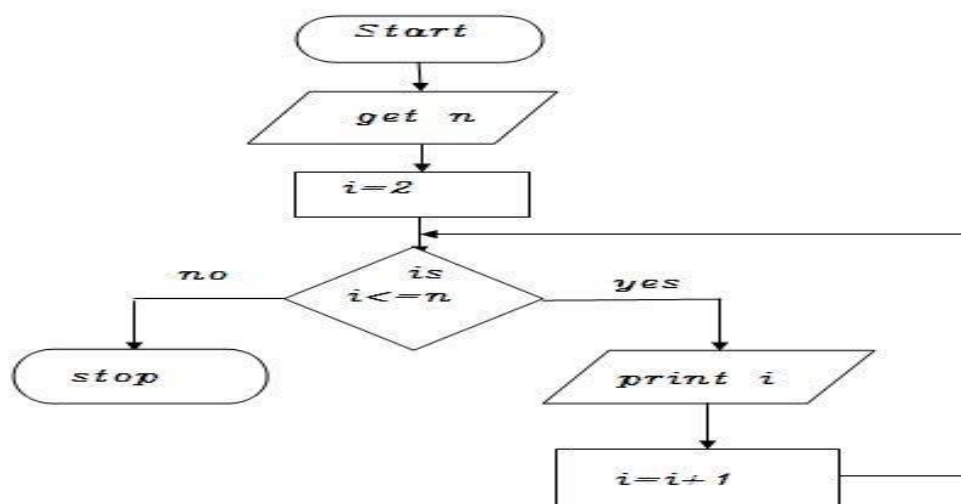
```

    PRINT i
    i=i+2
ENDWHILE
END

```

Write an algorithm to print n even numbers

Step 1: start
 step 2: get n value
 step 3: set initial value $i=2$
 step 4: check if($i \leq n$) goto step 5 else goto step8
 step 5: print i value
 step 6: increment i value by 2
 step 7: goto step 4
 step 8: stop



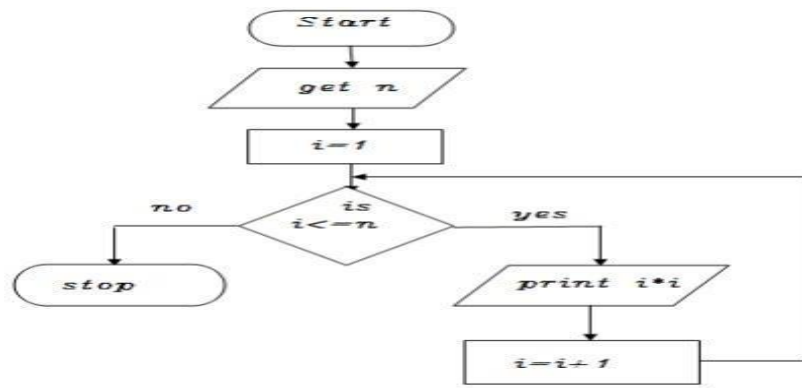
```

BEGIN
GET n
INITIALIZE i=2
WHILE(i<=n) DO
    PRINT i
    i=i+2
ENDWHILE
END

```

Write an algorithm to print squares of a number

Step 1: start
 step 2: get n value
 step 3: set initial value $i=1$
 step 4: check i value if($i \leq n$) goto step 5 else goto step8
 step 5: print $i*i$ value
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: stop

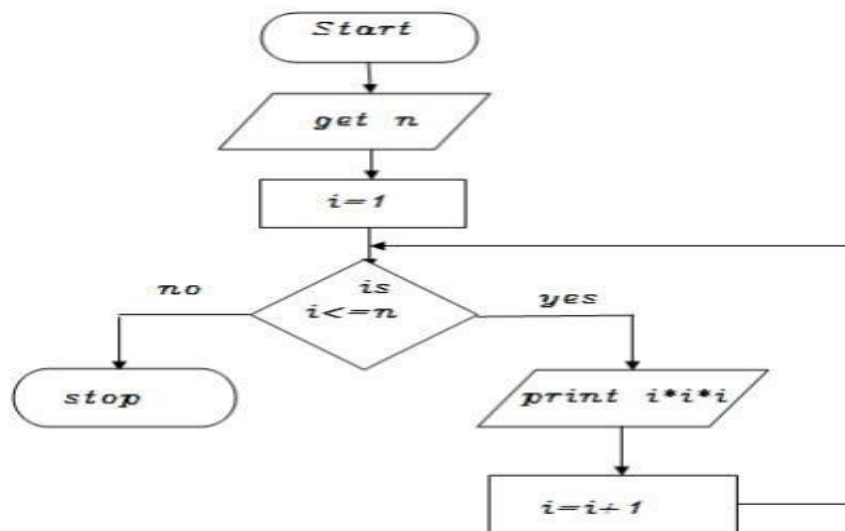


```

BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
    PRINT i*i
    i=i+1
ENDWHILE
END
  
```

Write an algorithm to print to print cubes of a number

Step 1: start
 step 2: get n value
 step 3: set initial value i=1
 step 4: check i value if(i<=n) goto step 5 else goto step 8
 step 5: print i*i *i value
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: stop



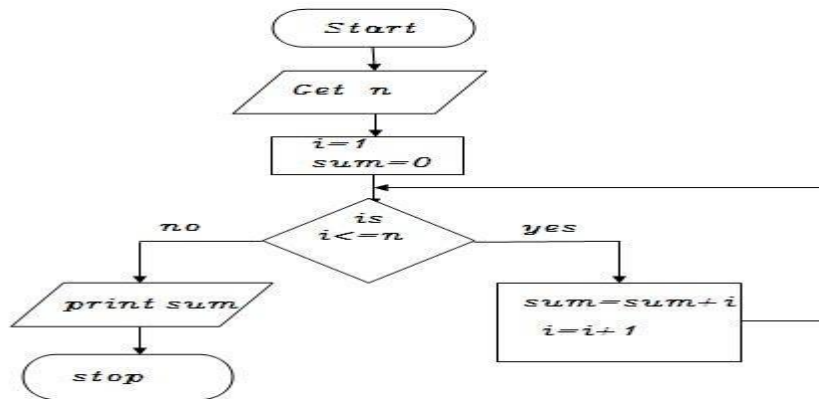
```

BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
    PRINT i*i*i
    i=i+1
ENDWHILE
END
  
```

Write an algorithm to find sum of a given number

Step 1: start

step 2: get n value
 step 3: set initial value $i=1$, $sum=0$
 Step 4: check i value if($i \leq n$) goto step 5 else goto step8
 step 5: calculate $sum=sum+i$
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: print sum value
 step 9: stop

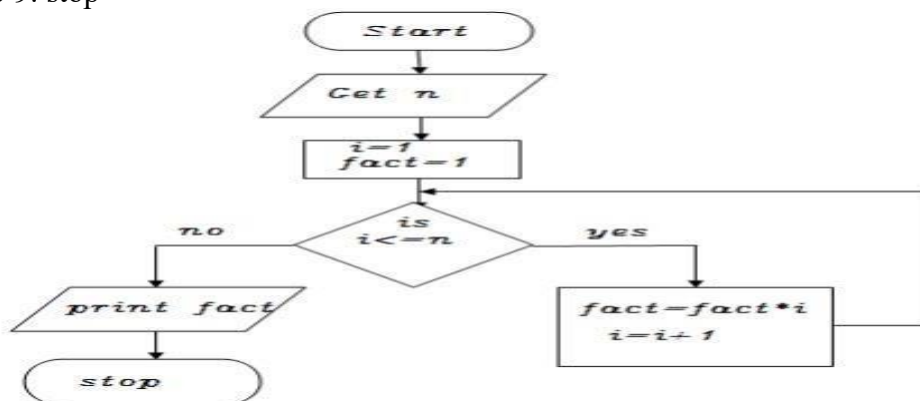


```

BEGIN
GET n
INITIALIZE i=1,sum=0
WHILE(i<=n) DO
    sum=sum+i
    i=i+1
ENDWHILE
PRINT sum
END
    
```

Write an algorithm to find factorial of a given number

Step 1: start
 step 2: get n value
 step 3: set initial value $i=1$, $fact=1$
 Step 4: check i value if($i \leq n$) goto step 5 else goto step8
 step 5: calculate $fact=fact*i$
 step 6: increment i value by 1
 step 7: goto step 4
 step 8: print fact value
 step 9: stop



```

BEGIN
GET n
INITIALIZE i=1,fact=1
    
```

```

WHILE(i<=n) DO
    fact=fact*i
    i=i+1
ENDWHILE
PRINT fact
END

```

Basic python programs:

Addition of two numbers

```

a=eval(input("enter first no"))
b=eval(input("enter second no"))
c=a+b
print("the sum is ",c)

```

Output

```

enter first no
5
enter second no
6
the sum is 11

```

Area of rectangle

```

l=eval(input("enter the length of rectangle"))
b=eval(input("enter the breath of rectangle"))
a=l*b
print(a)

```

Output

```

enter the length of rectangle 5
enter the breath of rectangle 6
30

```

Area & circumference of circle

```

r=eval(input("enter the radius of circle"))
a=3.14*r*r
c=2*3.14*r
print("the area of circle",a)
print("the circumference of circle",c)

```

output

```

enter the radius of circle4
the area of circle 50.24
the circumference of circle
25.12

```

Calculate simple interest

```

p=eval(input("enter principle amount"))
n=eval(input("enter no of years"))
r=eval(input("enter rate of interest"))
si=p*n*r/100
print("simple interest is",si)

```

Output

```

enter principle amount 5000
enter no of years 4
enter rate of interest6

```

simple interest is 1200.0

Calculate engineering cutoff

```
p=eval(input("enter physics marks"))
c=eval(input("enter chemistry marks"))
m=eval(input("enter maths marks"))
cutoff=(p/4+c/4+m/2)
print("cutoff =",cutoff)
```

Output

```
enter physics marks 100
enter chemistry marks 99
enter maths marks 96
cutoff = 97.75
```

Check voting eligibility

```
age=eval(input("enter ur age"))
If(age>=18):
print("eligible for voting")
else:
print("not eligible for voting")
```

output

```
Enter ur age
19
Eligible for voting
```

Find greatest of three numbers

```
a=eval(input("enter the value of a"))
b=eval(input("enter the value of b"))
c=eval(input("enter the value of c"))
if(a>b):
if(a>c):
print("the greatest no is",a)
else:
print("the greatest no is",c)
else:
if(b>c):
print("the greatest no is",b)
else:
print("the greatest no is",c)
```

output

```
enter the value of a 9
enter the value of b 1
enter the value of c 8
the greatest no is 9
```

Print n natural numbers

```
for i in range(1,5,1):
print(i)
```

Output

1 2 3 4

Print n odd numbers

```
for i in range(1,10,2):  
    print(i)
```

Output

1 3 5 7 9

Print n even numbers

```
for i in range(2,10,2):  
    print(i)
```

Output

2 4 6 8

Print squares of numbers

```
for i in range(1,5,1):  
    print(i*i)
```

Output

1 4 9 16

Print squares of numbers

```
for i in range(1,5,1):  
    print(i*i*i)
```

Output

1 8 27 64

Print n natural numbers

```
i=1  
while(i<=5):  
    print(i)  
    i=i+1
```

Output

1
2
3
4
5

Print n odd numbers

```
i=2  
while(i<=10):  
    print(i)  
    i=i+2
```

Output

2
4
6
8
10

Print n even numbers

```
i=1  
while(i<=10):  
    print(i)  
    i=i+2
```

Output

```
1  
3  
5  
7  
9
```

Print n squares of numbers

```
i=1  
while(i<=5):  
    print(i*i)  
    i=i+1
```

Output

```
1  
4  
9  
16  
25
```

Print n cubes numbers

```
i=1  
while(i<=3):  
    print(i*i*i)  
    i=i+1
```

Output

```
1  
8  
27
```

find sum of n numbers

```
i=1  
sum=0  
while(i<=10):  
    sum=sum+i  
    i=i+1  
print(sum)
```

Output

```
55
```

factorial of n numbers/product of n numbers

```
i=1  
product=1  
while(i<=10):  
    product=product*i  
    i=i+1  
print(product)
```

Output

```
3628800
```

sum of n numbers

```
def add():
a=eval(input("enter a value"))
b=eval(input("enter b value"))
c=a+b
print("the sum is",c)
add()
```

Output

```
enter a value
6
enter b value
4
the sum is 10
```

area of rectangle using function

```
def area():
l=eval(input("enter the length of rectangle"))
b=eval(input("enter the breath of rectangle"))
a=l*b
print("the area of rectangle is",a)
area()
```

Output

```
enter the length of
rectangle 20
enter the breath of
rectangle 5
the area of rectangle is
100
```

swap two values of variables

```
def swap():
a=eval(input("enter a value"))
b=eval(input("enter b value"))
c=a
a=b
b=c
print("a=",a,"b=",b)
swap()
```

Output

```
enter a value3
enter b value5
a= 5 b= 3
```

check the no divisible by 5 or not

```
def div():
n=eval(input("enter n value"))
if(n%5==0):
print("the number is divisible by 5")
else:
print("the number not divisible by 5")
div()
```

Output

```
enter n value10
the number is divisible by
```

find remainder and quotient of given no

```
def reminder():
a=eval(input("enter a"))
b=eval(input("enter b"))
R=a%b
print("the remainder is",R)
def quotient():
a=eval(input("enter a"))
b=eval(input("enter b"))
Q=a/b
print("the remainder is",Q)
reminder()
quotient()
```

Output

```
enter a 6
enter b 3
the remainder is 0
enter a 8
enter b 4
the remainder is 2.0
```

convert the temperature

```
def ctof():
c=eval(input("enter temperature in centigrade"))
f=(1.8*c)+32
print("the temperature in Fahrenheit is",f)
def ftoc():
f=eval(input("enter temp in Fahrenheit"))
c=(f-32)/1.8
print("the temperature in centigrade is",c)
ctof()
ftoc()
```

Output

```
enter temperature in
centigrade 37
the temperature in
Fahrenheit is 98.6
enter temp in Fahrenheit
100
the temperature in
centigrade is 37.77
```

program for basic calculator

```
def add():
a=eval(input("enter a value"))
b=eval(input("enter b value"))
c=a+b
print("the sum is",c)
def sub():
a=eval(input("enter a value"))
b=eval(input("enter b value"))
c=a-b
```

```

print("the diff is",c)
def mul():
a=eval(input("enter a value"))
b=eval(input("enter b value"))
c=a*b
print("the mul is",c)
def div():
a=eval(input("enter a value"))
b=eval(input("enter b value"))
c=a/b
print("the div is",c)
add()
sub()
mul()
div()

```

Output

```

enter a value 10
enter b value 10
the sum is 20
enter a value 10
enter b value 10
the diff is 0
enter a value 10
enter b value 10
the mul is 100
enter a value 10
enter b value 10
the div is 1

```

ALGORITHMIC PROBLEM SOLVING

1. What is an algorithm?

Algorithm is an ordered sequence of finite, well defined, unambiguous instructions for completing a task. It is an English-like representation of the logic which is used to solve the problem. It is a step- by-step procedure for solving a task or a problem. The steps must be ordered, unambiguous and finite in number.

2. Write an algorithm to find minimum of 3 numbers in a list. ALGORITHM : Find Minimum of 3 numbers in a list

Step 1: Start

Step 2: Read the three numbers A, B, C

Step 3: Compare A and B.

If A is minimum, go to step 4 else go to step 5. Step 4: Compare A and C.

If A is minimum, output "A is minimum" else output "C is minimum". Go to step 6.

Step 5: Compare B and C.

If B is minimum, output "B is minimum" else output "C is minimum".

Step 6: Stop

3. List the building blocks of an algorithm.

The building blocks of an algorithm are

1. Statements
2. Sequence
3. Selection or Conditional
4. Repetition or Control flow

5. Functions

4. Define statement. List its types.

Statements are instructions in Python designed as components for algorithmic problem solving, rather than as one-to-one translations of the underlying machine language instruction set of the computer.

There are three types of high-level programming language statements Input/output statements make up one type of statement. An input statement collects a specific value from the user for a variable within the program. An output statement writes a message or the value of a program variable to the user's screen.

5. Write the pseudo code to calculate the sum and product of two numbers and display it.

INITIALIZE variables sum, product, number1, number2 of type real

PRINT "Input two numbers"

READ number1, number2

sum = number1 + number2

PRINT "The sum is ", sum

COMPUTE product = number1 * number2

PRINT "The Product is ", product

END program

6. How does flow of control work?

Control flow (or flow of control) is the order in which individual statements, instructions or function calls of an imperative program are executed or evaluated. A control flow statement is a statement in which execution results in a choice being made as to which of two or more paths to follow.

7. What is a function?

Functions are "self-contained" modules of code that accomplish a specific task. Functions usually "take in" data, process it, and "return" a result. Once a function is written, it can be used over and over and over again. Functions can be "called" from the inside of other functions.

8. Write the pseudo code to calculate the sum and product displaying the answer on the monitor screen.

INITIALIZE variables sum, product, number1, number2 of type real

PRINT "Input two numbers"

READ number1, number2 sum = number1 + number2

PRINT "The sum is ", sum

COMPUTE product = number1 * number2

PRINT "The Product is ", product

END program

9. Give the rules for writing Pseudo codes.

1. Write one statement per line.
2. Capitalize initial keywords.
3. Indent to show hierarchy.
4. End multiline structure.
5. Keep statements to be language independent.

10. Give the difference between flowchart and pseudo code.

Flowchart and Pseudo code are used to document and represent the algorithm. In other words, an algorithm can be represented using a flowchart or a pseudo code. Flowchart is a graphical

representation of the algorithm. Pseudo code is a readable, formally styled English like language representation of the algorithm.

11. Define a flowchart.

A flowchart is a diagrammatic representation of the logic for solving a task. A flowchart is drawn using boxes of different shapes with lines connecting them to show the flow of control. The purpose of drawing a flowchart is to make the logic of the program clearer in a visual form.

12. Give an example of iteration.

```
a = 0
for i from 1 to 3      // loop three times
{
a = a + I      // add the current value of i to a
}
print a // the number 6 is printed (0 + 1; 1 + 2; 3 + 3)
```

13. Write down the rules for preparing a flowchart.

While drawing a flowchart, some rules need to be followed—

- (1) A flowchart should have a start and end,
- (2) The direction of flow in a flowchart must be from top to bottom and left to right, and
- (3) The relevant symbols must be used while drawing a flowchart.

14. List the categories of Programming languages.

Programming languages are divided into the following categories:

Interpreted, Functional, Compiled, Procedural, Scripting, Markup, Logic-Based, Concurrent and Object-Oriented Programming Languages

15. Mention the characteristics of an algorithm.

1. Algorithm should be precise and unambiguous.
2. Instruction in an algorithm should not be repeated infinitely.
3. Ensure that the algorithm will ultimately terminate.
4. Algorithm should be written in sequence.
5. Algorithm should be written in normal English.
6. Desired result should be obtained only after the algorithm terminates.

16. Compare machine language, assembly language and high-level language.

Machine language is a collection of binary digits or bits that the computer reads and interprets. This language is not easily understandable by the human.

An assembly language directly controls the physical hardware. A program written in assembly language consists of a series of instructions mnemonics that correspond to a stream of executable instructions, when translated by an assembler can be loaded into memory and executed. The programs written in this language are not portable and the debugging process is also not very easy.

A high level language is much more abstract, that must be translated or compiled in to machine language. It is easily understandable and the programs are portable. Debugging the code is easy and the program written is not machine dependent.

17. What is the difference between algorithm and pseudo code?

An algorithm is a systematic logical approach used to solve problems in a computer while pseudo code is the statement in plain English that may be translated later to a programming language. Pseudo code is the intermediary between algorithm and program.

18. List out the simple steps to develop an algorithm. Algorithm development process consists of five major steps.

- Step 1: Obtain a description of the problem.
 Step 2: Analyze the problem.
 Step 3: Develop a high-level algorithm.
 Step 4: Refine the algorithm by adding more detail.
 Step 5: Review the algorithm.

19. Give the differences between recursion and iteration.

Recursion	Iteration
Function calls itself until the base condition is reached.	Repetition of process until the condition fails.
Only base condition (terminating condition) is specified.	It involves four steps: initialization, condition, execution and updation.
It keeps our code short and simple.	Iterative approach makes our code longer.
It is slower than iteration due to overhead of maintaining stack.	Iteration is faster.
It takes more memory than iteration due to overhead of maintaining stack.	Iteration takes less memory.

20. What are advantages and disadvantages of recursion?

Advantages	Disadvantages
Recursive functions make the code look clean and elegant.	Sometimes the logic behind recursion is hard to follow through.
A complex task can be broken down into simpler sub-problems using recursion.	Recursive calls are expensive (inefficient) as they take up a lot of memory and time.
Sequence generation is easier with recursion than using some nested iteration.	Recursive functions are hard to debug.

1. What is mean by problem solving?
2. List down the problem solving techniques?
3. Define algorithm?
4. What are the properties of algorithm?
5. List down the equalities of good algorithm?
6. Define statements?
7. Define state?
8. What is called control flow?
9. What is called sequence execution?
10. Define iteration?
11. What is mean by flow chart?
12. List down the basic symbols for drawing flowchart?
13. List down the rules for drawing the flowchart?
14. What are the advantages of flowchart?
15. What are the disadvantages of flowchart?
16. Define pseudo code?
17. List down the keywords used in writing pseudo code?
18. Mention the advantages of using pseudo code?
19. Mention the disadvantages of using pseudo code?
20. What are the ways available to represent algorithm?
21. Differentiate flowchart and pseudo code?
22. Differentiate algorithm and pseudo code?
23. What is programming language?
24. Mention the types of programming language?
25. What is mean by machine level language?

26. What are the advantages and disadvantages of machine level language?
27. What is high level programming language and mention its advantages?
28. What are the steps in algorithmic problem solving?
29. Write the algorithm for any example?
30. Draw the flow chart for any example?
31. Write pseudo code for any example?

1. What are the building blocks of an algorithm? Explain in detail.

The building blocks of algorithm are -Statements

-State

-Control flow

-Functions

Statements:

There are 3 types of statements:

-Input/Output Statement

-Assignment Statement

-Control Statement

State:

There are 3 types of state:

-Initial state

-Current state

-Final state

Control flow:

-Sequence

The sequence structure is the construct where one statement is executed after another -

Selection

The selection structure is the construct where statements can be executed or skipped depending on whether a condition evaluates to TRUE or FALSE. There are three selection structures in C:

1. IF
2. IF – ELSE
3. SWITCH

Repetition

The repetition structure is the construct where statements can be executed repeatedly until a condition evaluates to TRUE or FALSE. There are two repetition structures in C:

1. WHILE
2. FOR

Functions:

A function is a block of organized reusable code that is used to perform a single action.

2. Briefly describe iteration and recursion. Illustrate with an example.

ITERATION:

S.no.	Iteration	Recursion
1	The process is repeated until the condition fails.	The function calls itself until the base condition is satisfied.
2	It consumes less memory.	It consumes more memory
3	It is faster	It is slower
4	The code is long .	The code is short.
5	Tracing is easier if any problem occurs.	Tracing is difficult if any problem occurs.

Example: Iterative algorithm for factorial of a number

```

Step 1: Start
Step 2: Read number  $n$ 
Step 3: Call factorial( $n$ )
Step 4: Print factorial  $f$ 
Step 5: Stop

factorial( $n$ )
Step 1: Initialize  $f=1, i=1$ 
Step 2: Repeat step 2.1 and 2.2 until  $i \leq n$ 
    Step 2.1:  $f = f * i$ 
    Step 2.2: Increment  $i$  by 1 ( $i = i + 1$ )
Step 3: Return  $f$ 
    
```

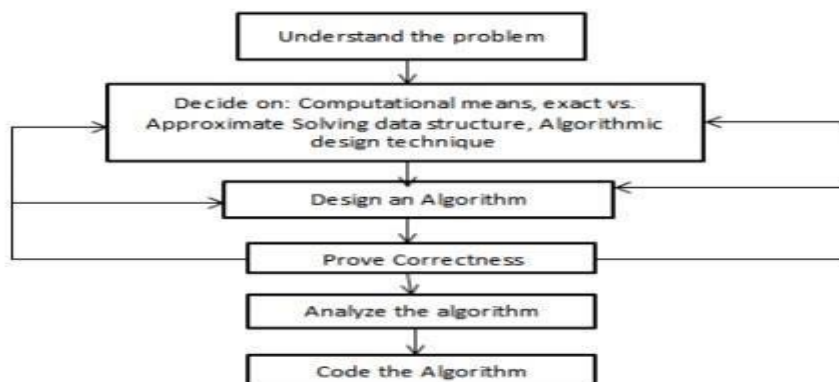
Example: Recursive algorithm for factorial of number

```

Step 1: Start
Step 2: Read number  $n$ 
Step 3: Call factorial( $n$ )
Step 4: Print factorial  $f$ 
Step 5: Stop

factorial( $n$ )
Step 1: If  $n == 1$  then return 1
Step 2: Else
     $f = n * \text{factorial}(n-1)$ 
Step 3: Return  $f$ 
    
```

3. Explain in detail Algorithmic problem solving.



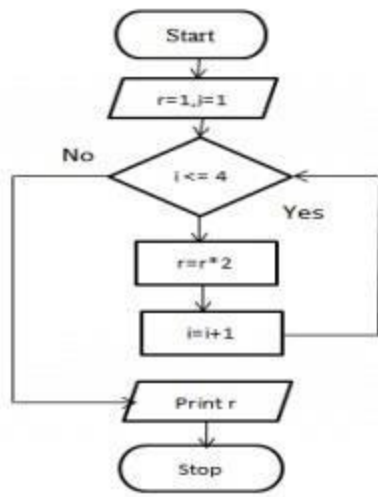
4. Write an algorithm and draw a flowchart to calculate 2^4 .

Algorithm:

```

Step 1: Start
Step 2: Initialize the value of result,  $r=1$ .
Step 3: Repeat step4 for 4 times
Step 4: calculate  $r=r*2$ 
Step 5: Print the value of  $r$ 
Step 6: Stop
    
```

Flowchart:



5. a) Describe pseudo code with its guidelines.

Pseudo code consists of short, readable and formally-styled English language used for explaining an algorithm. Pseudo code does not include details like variable declarations, subroutines etc.

Preparing a Pseudo Code

1. Pseudo code is written using structured English.
2. In a pseudo code, some terms are commonly used to represent the various actions. For example,
for inputting data the terms may be (INPUT, GET, READ),
for outputting data (OUTPUT, PRINT, DISPLAY),
for calculations (COMPUTE, CALCULATE),
for incrementing (INCREMENT),
in addition to words like ADD, SUBTRACT, INITIALIZE used for addition, subtraction, and initialization, respectively.
3. The control structures—sequence, selection, and iteration are also used while writing the pseudo code.
4. The sequence structure is simply a sequence of steps to be executed in linear order.
5. The selection constructs—if statement and case statement. In the if-statement, if the condition is true then the THEN part is executed otherwise the ELSE part is executed. There can be variations of the if-statement also, like there may not be any ELSE part or there may be nested ifs. The case statement is used where there are a number of conditions to be checked. In a case statement, depending on the value of the expression, one of the conditions is true, for which the corresponding statements are executed. If no match for the expression occurs, then the OTHERS option which is also the default option, is executed.
6. WHILE and FOR are the two iterative statements.

b) Give an example for pseudo code.

Pseudocode for finding maximum in a list:

```

BEGIN
SET numlist=[ ]
GET n
FOR i=1 to n
GET numlist elements
ENDFOR
SET maximum = numlist[0]
FOR i in numlist

```

```

IF (n > maximum)
maximum = n
ENDIF
ENDFOR
PRINT maximum
END
    
```

c) Write the pseudo code for Towers of Hanoi.

Pseudocode

```

START
Procedure Hanoi(disk, source, dest, aux)
IF disk == 0, THEN
move disk from source to dest
ELSE
Hanoi(disk - 1, source, aux, dest)
move disk from source to dest
Hanoi(disk - 1, aux, dest, source)
END IF
END Procedure
    
```

6. a) What is flowchart?

Flowchart is a diagrammatic representation of the logic for solving a task. A flowchart is drawn using boxes of different shapes with lines connecting them to show the flow of control. The purpose of drawing a flowchart is to make the logic of the program clearer in a visual form.

b) List down symbols and rules for writing flowchart.

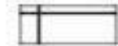
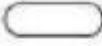
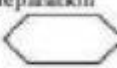
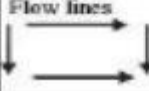
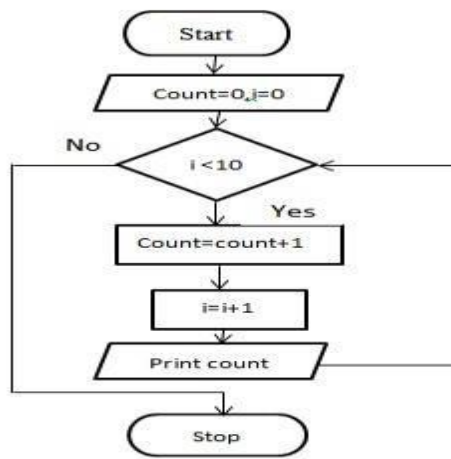
Process 	Alternate process 	Decision 	Data 	Predefined process 
Internal storage 	Document 	Multi document 	Terminator 	Preparation 
Manual input 	Manual operation 	Connector 	Off-page connector 	Card 
Punched tape 	Summing junction 	OR 	Collate 	Sort 
Extract 	Merge 	Stored data 	Delay 	Sequential access storage 
Magnetic disk 	Direct access storage 	Display 	Flow lines 	

Figure : Flow chart Symbols

c) Draw a flowchart to count and print from 1 to 10.



7. a) Write an algorithm and give the flowchart to find the net salary of an employee.

Algorithm:

Step 1: Start

Step 2 : Read the basic salary

Step 3 : IF the basic is greater than or equal to 4000 ELSE Goto Step 4

Step 3.1 : $DA = 0.32 * \text{basic}$ (Dearness Allowance)

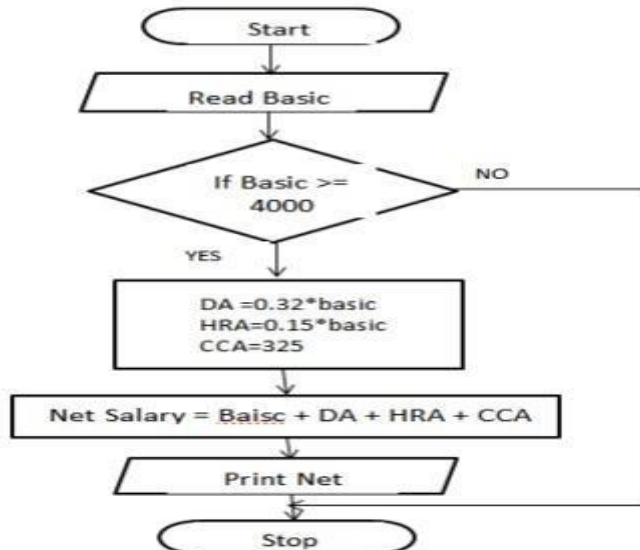
Step 3.2 : $HRA = 0.15 * \text{basic}$ (House Rent Allowance)

Step 3.3 : $CCA = 325$ (City Compensatory Allowance)

Step 3.4 : Net Salary basic + DA HRA + CCA

Step 4 : Print the Net Salary

Step 5 : Stop



b) Write an algorithm and give the pseudo code to guess an integer number in a range.

Algorithm:

step 1: Start the program

step 2: Read an 'n' number

step 3: Read an Guess number

step 4: if $\text{Guess} > n$; print "Your Guess too high" Step 5: elif $\text{Guess} < n$; print "Your Guess too low"

step 6: elif $\text{Guess} == n$; print "Good job"

Step 7: else print "Nope "

Step :8 Stop the program

Pseudocode:

```

BEGIN
READ n
READ Guess = 20
IF Guess > n
print "Your Guess too High" elif Guess < n
print "Your Guess too low" elif Guess == 20
print "Good Job"
ELSE
print "Nope"

```

8. a) Write an algorithm to insert a card in a list of sorted cards.

ALGORITHM:

```

Step 1: Start
Step 2: Declare the variables N, List[], I and X
Step 3: READ Number of element in sorted list as N
Step 4: SET i=0
Step 5: IF i<N THEN go to step 6 ELSE go to step 9
Step 6: READ Sorted list element as List[i]
Step 7: i=i+1
Step 8: go to step 5
Step 9: READ Element to be insert as X
Step 10: SET i=N-1
Step 11: IF i>0 AND X<List[i] THEN go to step 12 ELSE go to step 15
Step 13: i=i-1
Step 14: go to step 11
Step 15: List[i+1]=X
Step 16: Stop

```

b) Write an algorithm to find the minimum number in a list.

Algorithm:

```

Step 1 : Start
Step 2 : Initialize the value of minimum = 0
Step 3 : Enter the input number (n) of items in a list.
Step 4 : Get all the elements using for loop and store it in a list.
Step 5: Assign the first element in a list as minimum.
Step 6: Compare maximum with the first element in a list,n.
Step 7: Repeat step 8,9 until list becomes empty.
Step 8 : If n is less than the minimum
Step 9 : Assign minimum = n
Step 10 : Display minimum

```

Pseudocode:

```

BEGIN
SET numlist=[ ]
GET n
FOR i=1 to n
GET numlist elements
ENDFOR
SET minimum = numlist[0]
FOR i in numlist
IF (n < minimum)
minimum = n
ENDIF
ENDFOR
PRINT minimum
END

```

BART B:

1. Explain in detail about problem solving techniques?
2. Explain in detail about building blocks of algorithm?
3. Discuss the symbols and rules for drawing flowchart with the example?
4. Explain in detail about programming language?
5. Discuss briefly about algorithmic problem solving?
6. Write algorithm, pseudo code and flow chart for any example?
7. Explain in detail about simple strategies for developing algorithms?